



# Environmental predictors of benthic meiofaunal densities in the abyssal and hadal zones of the North Pacific: a combined Gradient Forest and GLMM approach

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**Abstract** The northern North Pacific forms a heterogeneous deep-sea environment with dynamic currents, distinct water masses, and complex bathymetry, including the Kuril–Kamchatka (KKT) and Aleutian Trenches (AT). As a climate-sensitive region, warming and sea-ice loss alter surface productivity and organic matter flux, potentially affecting deep-sea benthic communities. We quantified metazoan meiofaunal densities across abyssal and hadal zones of the KKT and AT, integrating samples from the 2022 AleutBio (SO293) expedition with legacy datasets.

Gradient-Forest modeling captured broad productivity patterns across higher taxonomic levels, while generalized linear mixed effects models (GLMMs) quantified taxon-specific responses to environmental factors. Meiofaunal densities varied substantially, peaking on the abyssal KKT plain and reaching minima in the hadal AT. Gradient -Forest analysis identified surface chlorophyll a, bottom-water temperature, and monthly primary production as primary drivers of regional density patterns. However, taxon-specific GLMMs revealed that once spatial effects and over-dispersion were controlled, bottom-water chemistry and hydrography exerted the most consistent control on local abundances. These results demonstrate a two-tier ecological control: surface-derived food availability governs meiofaunal densities at regional

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scales, whereas bottom-water chemistry, hydrographic regimes, and grain size fine-tune local distributions. This study provides critical baseline data for understudied deep-sea regions and informs predictions of meiofaunal responses to climate-driven changes.

**Keywords** Aleutian Trench · Kuril Kamchatka Trench · Environmental gradients · Climate change · Deep sea

## Introduction

The deep sea, here defined as areas beyond the continental shelf break at >200 m depth (Tyler et al., 2016), encompasses diverse environments characterized by unique topographies, seabed properties, water-mass characteristics, and energy availability. These conditions influence the distribution and composition of life across various geomorphic features including vast abyssal plains, steep continental slopes and ridges, as well as incised canyons and hadal trenches that extend to the ocean's greatest depth (Ramirez-Llodra et al., 2010). In these settings, the resident biota depends heavily on organic matter produced in the euphotic zone and subsequently transported to the deep sea as the primary food resource (i.e., allochthonous inputs from external sources rather than autochthonous, locally produced material). However, only a fraction of the organic material produced in surface waters reaches the seafloor (Lutz et al., 2002), resulting in decreasing faunal densities and biomass with increasing depth (Rex et al., 2006; Wei et al., 2010). Yet, the strength of this decline is not uniform across different size classes. Specifically, meiofaunal communities experience a less severe decrease in densities compared to larger (macro- and megafaunal) organisms, indicating shifting ecological significance toward smaller biota with lower energy demands at greater depths (Thiel, 1975, 1979; Rex et al., 2006; Wei et al., 2010; Zeppilli et al., 2018).

Meiofauna, i.e., minute benthic eukaryotes including metazoans, protists and fungi that pass through a 1 mm sieve but are retained by a lower mesh size of 32 µm (Mare, 1942; Higgins & Thiel, 1988), play an integral role in deep-sea ecosystems (Zeppilli et al., 2018; Schmidt-Rhaesa, 2020; Martínez et al., 2025). Their ecological relevance is

highlighted by their contributions to ecosystem processes through their wide distribution, high abundance, diversity, and rapid generation times (Woodward, 2010; Schratzberger & Ingels, 2018). As essential components of the food web, meiofauna enhances nutrient cycling and sediment bioturbation, improving nutrient exchange within the sediment (Green & Chandler, 1994; Meadows & Meadows, 1994; Bonaglia & Nascimento, 2023). They also support sediment remineralization by fostering microbial activity and facilitating the breakdown and assimilation of organic matter (Findlay & Tenore, 1982; Moens et al., 2007; Ingels et al., 2023). Moreover, meiofauna serve as indicators for assessing the impacts of human activities on marine environments, including climate change effects (Coull & Chandler, 1992; Pusceddu et al., 2014; Zeppilli et al., 2015; Moens et al., 2022; Schratzberger et al., 2023). Yet, despite their significant ecological role, there exists a notable scarcity of comprehensive data on meiofaunal communities in certain regions, such as the deep North Pacific (Rosli et al., 2018) limiting our capacity to identify the environmental drivers shaping these communities and to predict their responses to environmental change.

The northern North Pacific is characterized by distinct water masses and strong current systems coupled with varied bathymetric features, including trenches, ridges, and seamounts (Warren & Owens, 1988; Harris et al., 2014). Prominent among these are the southwest-to-northeast directed Kuril Kamchatka Trench (KKT) and the east-to-west directed Aleutian Trench (AT) along the Aleutian Ridge. The Bering Strait, a shallow marine passage, constrains faunal migration from and into the Arctic (Lee et al., 2020; Lemenkova, 2021). This area is sensitive to climate perturbations, particularly in the south-eastern Bering Sea, which has undergone demonstrated ocean warming and changes in sea-ice coverage with subsequent effects on surface productivity and marine food webs (Coyle et al., 2007; Brown et al., 2011; Hunt et al., 2022; Overland et al., 2024). Such changes in surface-water conditions and the frequency and magnitude of organic-matter export to the deep sea can also profoundly affect deep-sea floor organisms (Smith et al., 2013). These environmental transitions make the North Pacific particularly suitable for investigating meiofaunal responses across multiple gradients of food availability, sediment characteristics,

and hydrographic variables, providing essential benchmarks for detecting and forecasting ecological changes in response to climate variability.

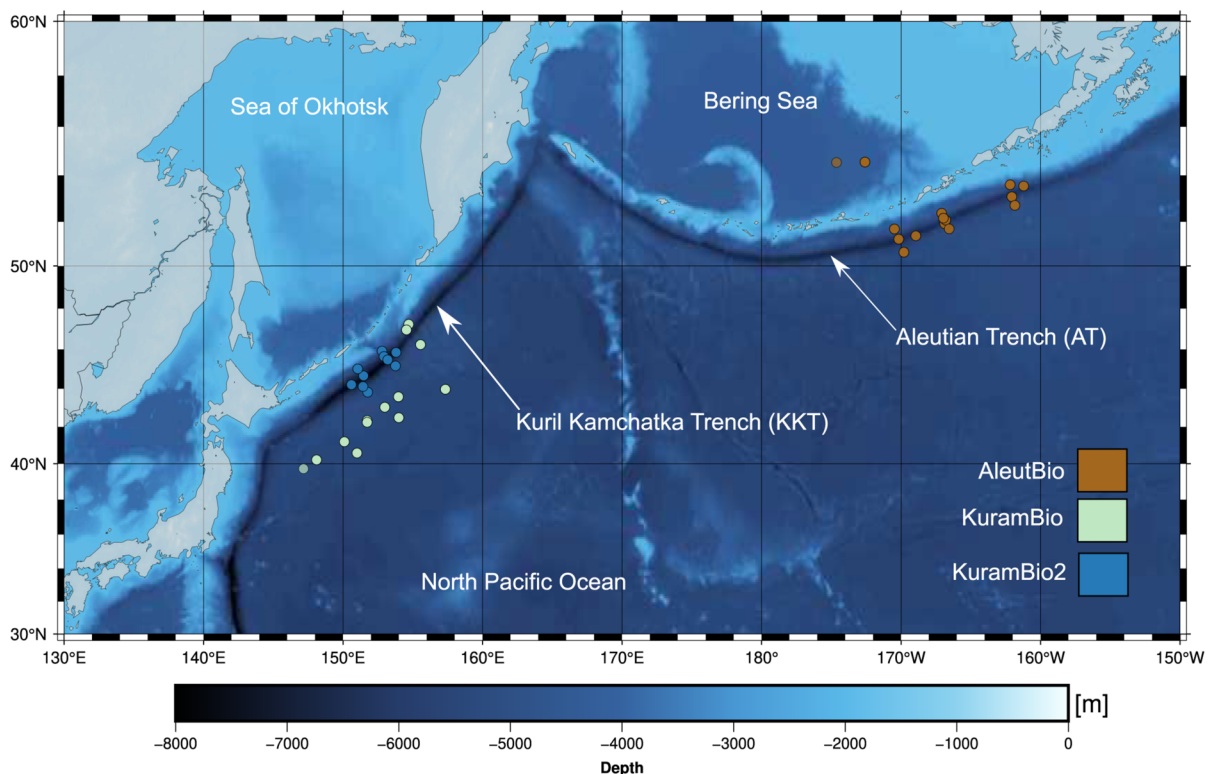
Incorporating new samples from the eastern AT (Brandt, 2022; Bonk et al., 2025), this study extends upon existing meiofaunal data from prior investigations in the KKT (Schmidt & Martínez Arbizu, 2015; Schmidt et al., 2019) to identify the environmental drivers affecting densities of metazoan meiofauna at higher taxonomic levels across these abyssal and hadal areas. Earlier studies identified seafloor particulate organic carbon (POC) flux and sediment characteristics, among others, as key determinants of meiofaunal densities (Pfannkuche, 1985; Pfannkuche & Thiel, 1987; Vanreusel et al., 1995; Hoste et al., 2007; Ostmann & Martínez Arbizu, 2018; Ingels et al., 2023). Building on this foundation, we aimed to (1) identify key environmental predictors of meiofaunal densities across bathymetric and geographic gradients; (2) quantify the relative importance of these predictors

and their taxon-specific effects using Gradient Forest and generalized linear mixed effects models (Poisson-GLMM); and (3) assess community thresholds along environmental gradients to elucidate the key drivers of deep-sea biodiversity patterns. This approach provides a baseline for predicting how ongoing changes in surface productivity and deep-water conditions may affect benthic communities in the northern North Pacific.

## Material and methods

### Study area

The study area covers two trench systems in the northern Pacific: the KKT adjacent to Japan, the Kuril Islands, and Kamchatka Peninsula, and the AT, which spans over 3,000 km from Siberia to the Alaska Peninsula (Fig. 1). The KKT is formed by the subduction of the Pacific Plate beneath the Okhotsk Plate toward



**Fig. 1** Map of multicorer sampling stations from SO223 (KuramBio, light blue), SO250 (KuramBio II, blue) and SO293 (AleutBio, orange) expeditions to the Kuril–Kamchatka and Aleutian trenches (northern North Pacific) and adjacent Bering Sea

the northwest (DeMets et al., 1994), while the AT results from the Pacific Plate subducting beneath the North American Plate (Lemenkova, 2021). The Subarctic North Pacific exhibits high nutrient levels with comparatively low chlorophyll concentrations (Harrison et al., 2004). The oceanography of the area is mainly influenced by the Western Subarctic Gyre to the west and the Alaska Gyre to the east, each characterized by distinct currents contributing to regional nutrient distribution. The Alaska Gyre, especially its northern edge marked by the strong westward-flowing Alaska Stream Current, influences water flow into the Bering Sea through passes in the Aleutian Islands, including Amchitka and Amukta passes and Near Strait (Stabeno et al., 1999; Zimmermann & Prescott, 2021). The Kamchatka Current defines the western edge of this gyre, directing water from the Siberian coast back toward the KKT (e.g., Khen et al., 2013). In the KKT, surface waters are further modulated by the contrasting influences of the cold, fresh Oyashio Current (from the Sea of Okhotsk) and the warmer, saltier Kuroshio-derived eddies and Kuril Current, while deep-water flows are dominated by the northward transport of the Lower Circumpolar Deep Water (LCDW) (Kawabe & Fujio, 2010). Some of this LCDW propagates into the AT via deep western-boundary currents (Johnson, 1998).

#### Sampling and taxonomic procedures

During the boreal summers of 2012, 2016, and 2022, samples were collected from the north-western and north-eastern Pacific Ocean in the course of three expeditions onboard RV *Sonne*: KuramBio (SO223), KuramBio II (SO250), and AleutBio (SO293) (Fig. 1, Online Resource 4). A multicorer (MUC) was used, employing cores with an inner diameter of 94 mm. Each core thereby provided a sampling surface area of 69.4 cm<sup>2</sup>. MUC sampling was carried out across 33 distinct sampling areas, with multiple deployments per area, resulting in a total of 66 stations. Previously published data from the KuramBio and KuramBio II expeditions can be found in Schmidt & Martínez Arbizu (2015) and Schmidt et al. (2019), respectively. Due to unit discrepancies between these earlier datasets and the most recent samples from SO293, a re-assessment of the raw data from Schmidt & Martínez Arbizu (2015) and Schmidt et al. (2019) was necessary. Specifically, meiofaunal

densities were calculated per core and harmonized to a standardized unit (number of individuals per 69.4 cm<sup>2</sup>). Furthermore, samples from the Bering Sea were excluded from the downstream analysis due to low sample coverage comprising eight cores, of which only two covered raster cells. This study includes data from a total of 66 individual stations comprising a total of 190 cores, with sampling depths ranging from 4,304 to 9,541 m (Bonk et al., 2026; Online Resource 4).

The sample protocol and processing of the MUC samples collected during SO293 are based on the methods described by Bonk et al. (2025). Briefly, three cores were analyzed at each station to evaluate the composition of meiofaunal assemblages. On board, the top 5 cm of each sediment core was preserved in buffered formalin (4%). The samples were then centrifuged and categorized into supra-specific taxa at the German Center for Marine Biodiversity Research (DZMB) in Wilhelmshaven, Germany.

Taxonomic resolution focused on supra-specific meiofaunal groups routinely distinguished in deep-sea meiofaunal studies. Nauplii, although representing larval stages of Copepoda rather than a separate taxon, were retained as an independent analytical category because they form a substantial and morphologically distinguishable component of benthic meiofaunal assemblages and may exhibit ecological responses distinct from copepodid and adult stages (e.g., Guidi-Guilvard et al., 2009; Guidi-Guilvard & Dallot, 2014; Ayón Dejo et al., 2023). This practice is commonly applied in meiofaunal ecology, particularly where nauplii contribute substantially to total abundance and can respond differently to environmental gradients. We therefore treat nauplii here as an operational ecological category rather than a formal taxonomic entity. Nevertheless, we acknowledge that this approach may underestimate total copepod representation when Copepoda and nauplii are considered separately. Conversely, typical macrofaunal taxa (e.g., Amphipoda, Isopoda, Polychaeta) were excluded, consistent with previous meiofaunal studies (e.g., Rowe et al., 1991; Berkenbusch et al., 2011).

#### Environmental layers

A set of environmental predictors was assembled to model meiofaunal distributions using GLMMs and Gradient Forests (Table 1). Benthic and water-column

**Table 1** Data sources for environmental predictors used to characterize meiobenthic communities in the northern North Pacific

| Acronym   | Parameter                                                                                  | Unit                       | Resolution | References                                  |
|-----------|--------------------------------------------------------------------------------------------|----------------------------|------------|---------------------------------------------|
| Chl       | Mean sea surface net primary productivity                                                  | g/m <sup>3</sup> /day      | 5 arcmin   | Bio-Oracle 3 Assis et al. (2024)            |
| SWS       | Mean sea water current velocity at maximum bottom depth                                    | m/s                        | 5 arcmin   | Bio-Oracle 3 Assis et al. (2024)            |
| O2        | Mean mole concentration of dissolved molecular oxygen in sea water at maximum bottom depth | μmol/m <sup>3</sup>        | 5 arcmin   | Bio-Oracle 3 Assis et al. (2024)            |
| SO        | Mean salinity at the seafloor                                                              | PSU                        | 5 arcmin   | Bio-Oracle 3 Assis et al. (2024)            |
| Temp      | Mean sea water temperature at the seafloor                                                 | degrees Celsius            | 5 arcmin   | Bio-Oracle 3 Assis et al. (2024)            |
| PO4       | Mean mole concentration of phosphate in sea water at maximum bottom depth                  | μmol/m <sup>3</sup>        | 5 arcmin   | Bio-Oracle 3 Assis et al. (2024)            |
| NO3       | Mean mole concentration of nitrate in sea water at maximum bottom depth                    | μmol/m <sup>3</sup>        | 5 arcmin   | Bio-Oracle 3 Assis et al. (2024)            |
| Si        | Mean concentration of silicate in sea water at maximum bottom depth                        | μmol/m <sup>3</sup>        | 5 arcmin   | Bio-Oracle 3 Assis et al. (2024)            |
| bathy     | Mean bottom depth                                                                          | m                          | 5 arcmin   | Bio-Oracle 3 Assis et al. (2024)            |
| Lutz_POC  | Particulate organic carbon flux to the seafloor based on Lutz et al. (2007) algorithm      | mgC/m <sup>2</sup> /day    | 5 arcmin   | Lutz et al. (2007); Snelgrove et al. (2018) |
| VGPM_svi  | Seasonal index of mean monthly surface primary production from 1998 to 2014                | VGPM~SD/mean               | 5 arcmin   | Lutz et al. (2007); Snelgrove et al. (2018) |
| VGPM_mean | Mean monthly surface primary production from 1998 to 2014                                  | mgC/m <sup>2</sup> /day    | 5 arcmin   | Lutz et al. (2007); Snelgrove et al. (2018) |
| SCOC      | Sediment- community oxygen consumption as Log10 of SCOC                                    | mmol/cm <sup>2</sup> /year | In situ    | Stratmann et al. (2019)                     |
| GLOBSED   | Total Sediment Thickness                                                                   | m                          | 5 arcmin   | Straume et al. (2019)                       |
| GR        | Seafloor roughness using 100 km filter width, unmasked                                     | mGal                       | 2 arcmin   | Whittaker et al. (2008)                     |

variables were obtained from Bio-Oracle v3 (Assis et al., 2024), including bathymetry (bathy), seafloor chlorophyll (Chl), dissolved oxygen (O<sub>2</sub>), salinity (SO), nitrate (NO<sub>3</sub>), phosphate (PO<sub>4</sub>), silicate (Si), seabed temperature (Temp), and current velocity (SWS). Benthic-flux indicators were obtained from Snelgrove et al. (2018), including particulate organic-carbon flux (Lutz et al., 2007), Vertically Generalized Production Model (VGPM) metrics (mean and seasonal variability [svi]), and sediment-community oxygen consumption. Total sediment thickness (GlobSed) was incorporated based on Straume et al. (2019), and seafloor roughness (Gravrough) was derived from EarthByte's gravity anomaly dataset following Whittaker et al. (2008). The environmental layers are provided at a global spatial resolution of 5 arcmin (~0.083°; approximately 9.2 km at the equator), representing large-scale environmental conditions rather than fine-scale habitat characteristics. All

layers were harmonized onto a common spatial grid, with full metadata including units, resolution, and sources detailed in Table 1. In addition, point-based sediment grain-size data obtained from MUC sampling were compiled from the literature (Schmidt & Martínez Arbizu, 2015; Schmidt et al., 2019; Sindlev et al., 2025), providing in situ measurements of benthic substrate characteristics.

#### Gradient Forest analysis

Species abundances are given as densities per sample area of 69.4 cm<sup>2</sup>. When more than one core was recovered from a single station, specimen counts were summed and then averaged and scaled to give density per core surface area (expressed per 69.4 cm<sup>2</sup>). We ran our Gradient-Forest analysis in R using the *gradient-Forest* package (Ellis et al., 2012) on a single combined data frame of the 15 environmental predictors and

taxon densities. We set ‘mtry=3’ to randomly sample three predictors at each split, and grew 2500 trees per taxon (‘ntree=2500’) to ensure stable importance estimates. Collinearity among predictors was addressed using conditional permutation importance (Strobl et al., 2008), which permutes predictors only within data blocks defined by splits of correlated variables ( $|r| > 0.5$ ) by setting ‘corr.threshold=0.5,’ i.e., the default value, preserving their structural dependence. Finally, we allowed the algorithm to explore up to three splits per tree (maxLevel). The resulting models produced per-predictor weighted  $R^2$  ( $wR^2$ ) values for ranking importance, as well as cumulative importance curves to identify threshold-sensitive ranges along each environmental gradient. Unlike the coefficient of determination ( $R^2$ ) in conventional regression analyses,  $wR^2$  values represent weighted measures of predictor importance aggregated across species-specific random forest models and are therefore intended for relative comparison among predictors rather than assessment of absolute explanatory power.

#### Poisson-GLMM

We fitted generalized linear mixed-effects models (GLMMs) with a Poisson error distribution to assess how each environmental parameter influences meiofauna densities, while accounting for variation in sample area and spatial clustering. The predictors (depth and the focal parameter) were centered to zero mean and scaled to unit variance to facilitate model convergence and interpretation. For each focal taxon–parameter combination, we built a model with taxon counts as the response and the focal predictor and sampling depth as fixed effects and log(sampled sediment area) as an offset. This approach allowed us to estimate direct effects on densities (individuals/m<sup>2</sup>), while preserving the count data structure. We did not build a full model containing all predictors as the number of samples with unique values for the environmental predictors was restricted to the number of stations. For the random effect structure, we nested sampling stations ( $n=66$ ) within sampling areas ( $n=33$ ). We then added an observation-level random effect (OLRE) by creating a unique factor for each sample ( $n=190$ ) to control for overdispersion. Before modeling, records with missing values were omitted.

The primary model formula was specified as:

$$\begin{aligned} \text{Count} \sim & \text{offset}(\log(\text{sample\_area})) \\ & + \text{depth\_scaled} + \text{parameter\_scaled} \\ & + (1 | \text{area/station}) + (1 | \text{olre}) \end{aligned} \quad (1)$$

Models were fitted with `glmer()` from the *lme4* package (Bates et al., 2015), using a Poisson distribution and the “bobyqa” optimizer to ensure reliable convergence. We compared the two models via Akaike information criterion (AIC) and formally assessed overdispersion using the R *performance* package (Lüdtke et al., 2021) and a custom Pearson-chi-square-based overdispersion ratio (Zuur et al., 2009). Model assumptions (linearity, collinearity, random-effects structure, residual distribution) were further evaluated using the respective plots and functions.

To quantify the proportion of variance explained by each fixed effect, we computed semi-partial  $R^2$  statistics with the *partR2* package (Stoffel et al., 2021), performing 99 parametric bootstraps for confidence intervals. We extracted overall marginal  $R^2$ , the inclusive  $R^2$  attributable to each predictor ( $IR^2$ ), and the associated structure-component (SC) and beta-weight (BW) metrics. We used  $IR^2$  and BW to interpret the importance of each parameter, and the respective strength and direction of each effect.

In *partR2*,  $IR^2$  for a predictor  $j$  is calculated as

$$IR_j^2 = (SC_j)^2 \times R_{\text{total}}^2 \quad (2)$$

where  $SC_j$  is the *structure coefficient*, i.e., the Pearson correlation between the centered-and-scaled predictor  $x_j$  and the model’s fitted values.  $R_{\text{total}}^2$  is the marginal  $R^2$  of the full model, i.e., the variance explained by all fixed effects.

To quantify and compare the strength and direction of individual environmental effects, we extracted the beta-weight (BW) for every centered-and-scaled predictor from the final Poisson-GLMMs. A BW is simply the model’s fixed-effect slope after all continuous covariates have been standardized to zero mean and unit variance; it therefore represents the expected change in the density and hence the multiplicative change in the expected organism abundance associated with a one-standard-deviation increase in that predictor. Because all predictors share the same scale, BW values can be interpreted as standardized effect sizes (Burnham & Anderson, 2002): larger absolute BW

values denote stronger direct effects, while the sign (positive or negative) indicates the direction of the relationship. We chose BW as an additional measure of “individual predictor importance”, because it conveys both effect magnitude and direction, and remains directly comparable across taxa, sites, and models. For each BW we report bootstrapped 95% confidence intervals using 99 bootstraps to estimate uncertainty and to identify significant effects.

### Compositional data analysis

Grain-size composition was quantified as the percentage of sand, mud (combined silt and clay fraction), and gravel (Schmidt & Martínez Arbizu, 2015; Schmidt et al., 2019; Sindlev et al., 2025). Because these components are constrained to sum to 100%, they constitute compositional data and cannot be analyzed as independent variables, as an increase in one component necessarily results in a decrease in at least one of the others. Therefore, we applied compositional data analysis (Aitchison, 1986) to transform grain-size measurements into orthogonal, i.e., uncorrelated and isometric log-ratio (ILR) components (Egozcue et al., 2003). These components describe the relative balance between percentages of sand and mud vs. gravel (ILR1) and sand vs. mud (ILR2). Components were created using the package *compositions* (van den Boogart et al., 2025). The effect of sediment grain size on meiofaunal densities was then assessed using the same Poisson GLMM with offset described above, with ILR1 and ILR2 included as fixed effects. Owing to incomplete sediment data, this analysis was based on a reduced dataset. In total, 52 samples (27.4%) from 17 stations (25.8%) were excluded because grain-size measurements were unavailable or were erroneous (Online Resource 7). Acari were not analyzed due to insufficient data.

## Results

### Meiofaunal abundance and taxonomic composition

The dataset comprised a total of 647,202 meiofaunal individuals sampled at 66 stations in the KKT and AT. Benthic densities varied substantially across the investigated regions (Fig. 2, Online Resources 1&2). The highest densities were recorded on the abyssal

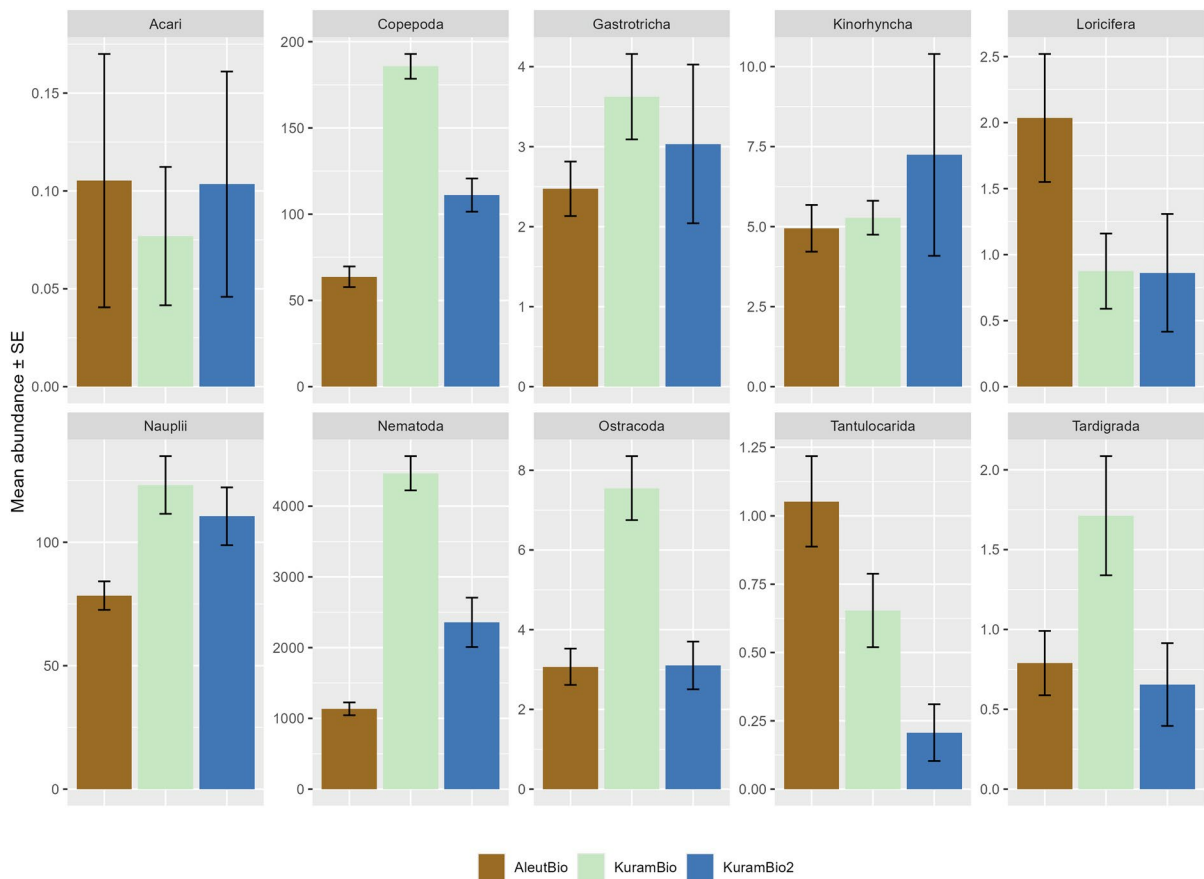
plain of the KKT, ranging from 37 to 1,589 individuals per 10 cm<sup>2</sup> ( $492.9 \pm 379.0$ , mean  $\pm$  SD). In contrast, the lowest densities occurred in the hadal zone of the AT, with values between 45 and 315 individuals per 10 cm<sup>2</sup> ( $178.5 \pm 108.8$ ). Intermediate densities were observed in the hadal KKT ( $82\text{--}1,115$  ind./10 cm<sup>2</sup>;  $399.6 \pm 281.3$ ) and the abyssal AT ( $107\text{--}478$  ind./10 cm<sup>2</sup>;  $249.4 \pm 203.9$ ).

The meiofaunal communities were dominated by Nematoda (92.2%), followed by Copepoda (4.1%) and nauplii (3.2%). These taxa were present in all samples and reached their highest densities on the abyssal KKT plain. Other taxa displayed more localized peaks: Kinorhyncha were most abundant in the hadal KKT, while Loricifera reached their maximum densities on the abyssal AT. Less abundant groups such as Kinorhyncha, Loricifera, Ostracoda, Tantulocarida, and Gastrotricha were detected at 8 (Tantulocarida) to 39 (Ostracoda) stations and consistently occurred at densities below 5 individuals per 10 cm<sup>2</sup>. Nevertheless, these taxa were retained for further analysis due to their consistent presence across multiple stations. In contrast, Acari were encountered only sporadically and in extremely low numbers, and were therefore excluded from subsequent analyses.

### Gradient Forest

The weighted-R<sup>2</sup> ranking (Fig. 3) shows that only few predictors account for most of the variance in meiofaunal composition. Mean surface chlorophyll-a (Chl) tops the list ( $wR^2 \approx 0.05$ ), followed closely by mean potential temperature at the seafloor (Temp), mean monthly surface primary production (VGPM\_mean), the seasonal variability of that production (VGPM\_svi), and mean salinity (SO). Together, these five variables explain roughly one-third of the cumulative importance, indicating that (i) pelagic productivity proxies, i.e., chlorophyll and VGPM metrics, form a strong trophic gradient, and (ii) hydrographic properties, bottom-water temperature and salinity, form an independent physical gradient that further structures the assemblage.

The next set of predictors comprises dissolved nutrients (phosphate, nitrate, and silicate) and dissolved oxygen as well as global sediment thickness, all with intermediate  $wR^2$  values (0.02–0.03). Their coherent positioning suggests a second environmental axis linked to nutrient availability, but their lower



**Fig. 2** Bar plots of mean abundances ( $\pm$  standard error, SE) of meiofauna taxa collected in different areas during the KuramBio & KuramBio II (SO223, SO250) and AleutBio (SO293) expeditions to the Kuril Kamchatka Trench, and Aleutian Trench

scores imply that this nutrient gradient is less influential for the taxonomic patterns observed than the primary productivity and water-mass characteristics. Finally, gravity roughness, SCOC and Lutz-POC rank lowest ( $wR^2 < 0.02$ ), confirming that large-scale model parameters played a comparatively minor role in shaping the broad-scale distribution of taxa in our data set.

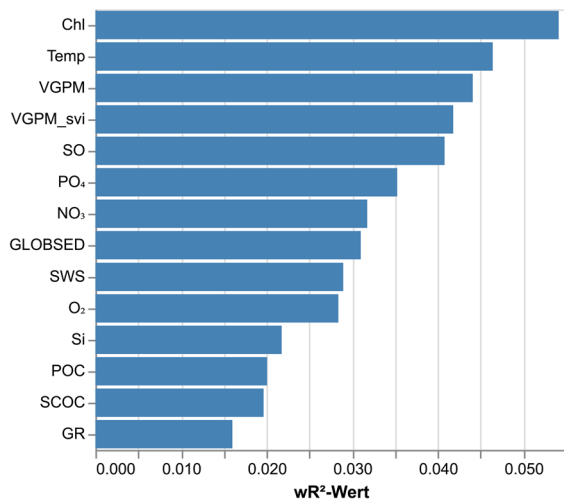
Because  $wR^2$  represents the average variance in species density that each predictor explains across all taxa, the steep drop-off after the top five reinforces the conclusion that meiofaunal communities in the study area are primarily associated with two orthogonal drivers: a productivity gradient (food supply) and a hydrographic gradient (temperature–salinity water mass), while nutrient concentrations contribute secondarily and sediment characteristics only marginally.

The cumulative importance curves for the six top ranking parameters (Fig. 4, see also Online Resource 2) confirm that productivity proxies (Chl, VGPM metrics) and hydrographic parameters (Temp, SO) define discrete ecological thresholds rather than broad linear trends: most taxa respond strongly within limited segments of each gradient, underscoring the presence of two principal, nonlinear environmental axes—food availability and characteristics—that structure meiofaunal assemblages in the study area.

#### Generalized linear mixed-effects models—GLMM

##### *Inclusive $R^2$ —Explanatory power of environmental predictors*

Figure 5 displays the heat-map of  $IR^2$  values, quantifying the unique fraction of variance that each



**Fig. 3** Gradient-Forest analysis of meiofaunal taxa. Weighted  $R^2$  values for each environmental variable taxon, representing the proportion of variance in taxon abundance explained by the model. Abbreviations: Chl—surface primary productivity; GLOBSED—Total Sediment Thickness; GR—Seafloor roughness;  $\text{NO}_3$ —Nitrate;  $\text{O}_2$ —dissolved oxygen concentration;  $\text{PO}_4$ —Phosphate; POC—Particulate organic carbon flux to the seafloor based on Lutz et al. (2007); SCOC—Sediment-community oxygen consumption; Si—Silicate; SO—salinity; Temp—Bottom temperature; SWS—sea-water current velocity; VGPM—monthly surface primary production; VGPM\_svi—Seasonal index of mean monthly surface primary production

predictor adds to the Poisson-GLMM for every meiofaunal group. Across most taxa, two coherent groups of predictor variables emerge as consistently high contributors to explained variability, as indicated by contiguous areas of elevated  $\text{IR}^2$  values in the variable heatmap. The first group comprises chemical parameters, primarily nitrate, oxygen and phosphate, with silicate contributing to a lesser extent. In Copepoda and Nematoda nitrate alone accounts for 15–18% of the total deviance, closely followed by oxygen (14–17%) and phosphate (11–13%). These same nutrients contribute modest yet detectable shares in nauplii, Loricifera and Tantulocarida, whereas their influence is negligible in Kinorhyncha, Gastrotricha, and Acari.

The second high-influence cluster comprises hydrographic descriptors. Bottom-water temperature and salinity each supply roughly 9–15% additional  $\text{IR}^2$  in Copepoda and Nematoda, and reach similar levels in Loricifera and Tantulocarida. Seawater current velocity (SWS) mirrors salinity for taxa that inhabit coarser sediments, contributing a further

5–10% for Copepoda and Ostracoda. Together, chemistry and hydrography account for more than half of the cumulative  $\text{IR}^2$  in all widespread taxa and dominate the aggregated community response.

Nutrient proxies have a narrower, taxon-specific footprint. Kinorhyncha displays pronounced sensitivities to surface chlorophyll-a ( $\text{IR}^2 \approx 0.11$ ) and mean primary production (0.10), with a smaller but consistent contribution from fresh-aggregate flux (0.05) and benthic oxygen demand (0.05), indicating reliance on steady phytodetrital input. In contrast, Tardigrada responds most strongly to production variability (VGPM\_svi, 0.04) and POC-flux pulses (0.07), and reduces markedly in highly rough sediments, where grain micro-relief alone explains 13% of its deviance—the highest single-predictor share for this group. Ostracoda also tracks organic matter delivery, with Lutz-POC reflecting 12% unique variance and density adding another 5%; productivity variables are otherwise minor for this taxon.

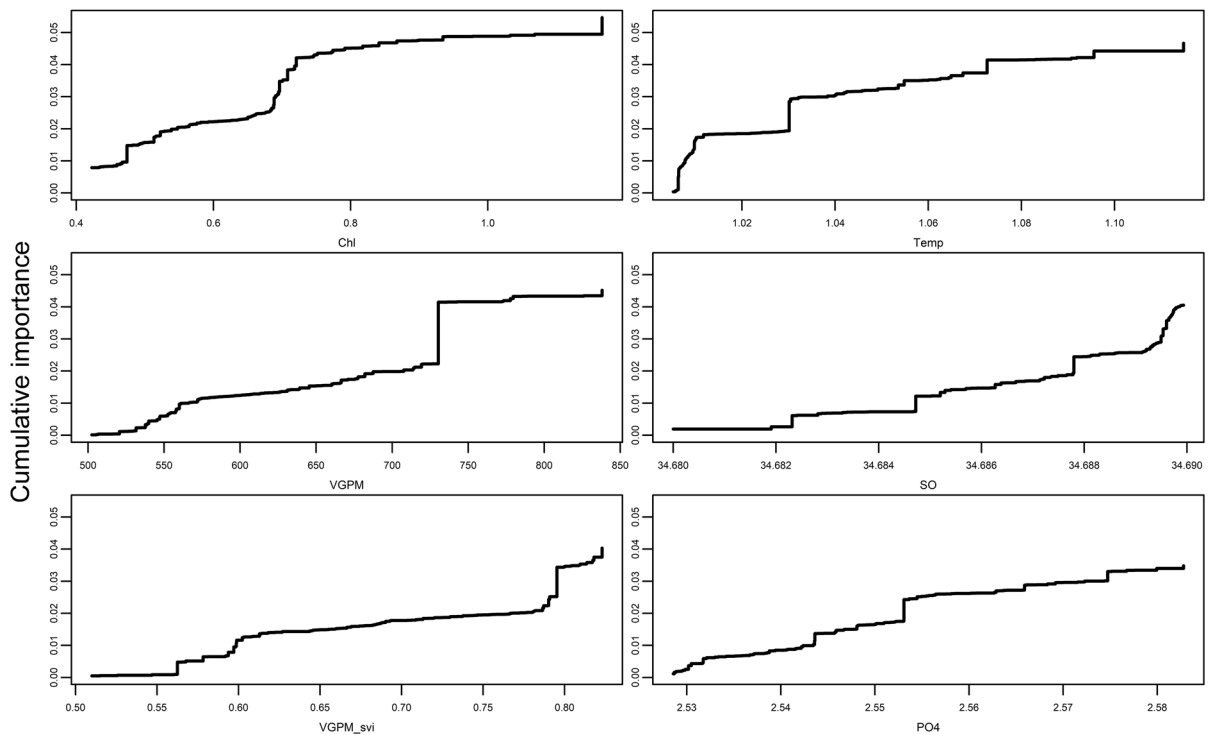
Sediment characteristics emerge as secondary factors. Aside from the roughness effect on Tardigrada and a moderate SCOC contribution to Kinorhyncha ( $\text{IR}^2 \approx 0.05$ ), habitat texture rarely exceeds three per cent in any model. Acari are exceptional in that no predictor explains more than two per cent of their variation, suggesting either strong stochasticity or dependence on unmeasured micro-scale cues.

Taken together, the  $\text{IR}^2$  patterns show that dissolved nutrients and redox state, coupled with properties, provide the most consistent explanatory power across the meiofaunal assemblage, whereas productivity signals refine the response of specialist deposit feeders and episodic organic influx shapes only a handful of groups. Sediment structure exerts taxon-limited but occasionally pronounced effects, underscoring the hierarchical organization of drivers from basin-scale chemistry to patch-scale habitat complexity.

#### Beta-weight (BW)

The standardized slope analysis, evaluated with 95% confidence intervals, singles out two environmental axes that represent the clearest, statistically supported influence on the meiofaunal assemblage: a nutrient oxygen gradient and a hydrographic gradient.

Along the nutrient oxygen axis, four of the dominant groups—Copepoda, nauplii, Nematoda,



**Fig. 4** Cumulative importance of the six most influential environmental predictors (Chl, Temp, VGPM, VGPM\_svi, SO, PO<sub>4</sub>) across all taxa. Large steps in the curves indicate strong changes in community composition along the respective environmental gradient. Abbreviations: Chl—surface pri-

mary productivity; PO<sub>4</sub>—Phosphate; SO: salinity; Temp—Bottom temperature; VGPM—monthly surface primary production; VGPM\_svi—Seasonal index of mean monthly surface primary production

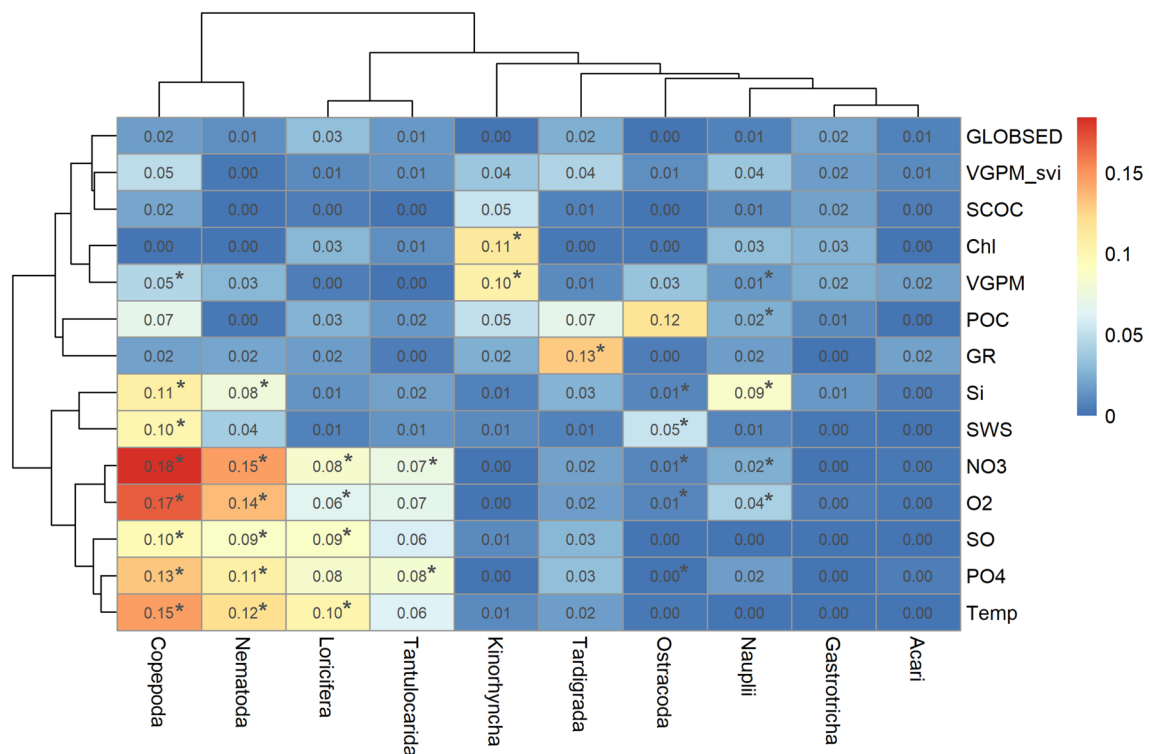
and Ostracoda—respond in the same direction. Their densities fall as nitrate, phosphate and silicate rise ( $\beta_w = -0.15$  to  $-0.48$ ,  $CI < 0$ ) and rise as dissolved oxygen increases ( $\beta_w = +0.19$  to  $+0.72$ ,  $CI > 0$ ) (Online Resource 5). Loricifera exhibits the reverse pattern, with significant positive slopes for nitrate ( $\beta_w = +0.89$ ,  $CI > 0$ ) and a marked negative slope for oxygen ( $\beta_w = -0.77$ ,  $CI < 0$ ), indicating a preference for low-oxygen, nutrient-rich pockets. Tantulocarida overlap partially with this niche, increasing with both nitrate and phosphate ( $\beta_w \approx +0.39$  and  $+0.42$ ,  $CI > 0$ ). No other taxa show significant nutrient or oxygen effects.

The hydrographic gradient reinforces this division. Bottom-water temperature and salinity increase Copepoda and Nematoda (salinity  $\beta_w \approx +0.33$ ,  $CI > 0$ ; temperature  $\beta_w \approx -0.37$ ,  $CI < 0$  when expressed as deviation from the mean), yet have the opposite effect on Loricifera (temperature  $\beta_w = +0.99$ ,  $CI > 0$ ; salinity  $\beta_w = -0.90$ ,  $CI < 0$ ).

Seawater density ( $\sigma_\theta$ ) is a secondary but significant positive factor for Copepoda and Ostracoda ( $\beta_w \approx +0.24$  to  $+0.25$ ,  $CI > 0$ ).

Nutrient variables exert significant control on only a few specialists. Kinorhyncha responded positively to steady phytodetritus: both surface chlorophyll-a and mean primary production show significant positive slopes ( $\beta_w \approx +0.40$ ,  $CI > 0$ ), and fresh-aggregate flux is similarly favorable ( $\beta_w \approx +0.41$ ,  $CI > 0$ ). Nauplii also increased with aggregate flux ( $\beta_w = +0.24$ ,  $CI > 0$ ). A single productivity metric, VGPM\_svi, is significant for Copepoda ( $\beta_w = +0.27$ ,  $CI > 0$ ), suggesting opportunistic exploitation of patchy food inputs; it remains non-significant elsewhere.

Sediment characteristics play only a minor role. Seafloor roughness is the lone habitat feature with a robust effect, strongly depressing Tardigrada ( $\beta_w = -0.82$ ,  $CI < 0$ ); all other sediment variables, including sediment-community oxygen consumption



**Fig. 5** Heatmap of the inclusive  $R^2$  ( $IR^2$ ) values per taxon-parameter derived from the GLMM. Higher values indicate a stronger effect of the environmental parameter alone when depth is kept at a mean value. Asterisks indicate significant standardized slopes. Abbreviations: Chl–surface primary productivity; GLOBSED–Total Sediment Thickness; GR–Seafloor roughness;  $NO_3$ –Nitrate;  $O_2$ –dissolved oxygen concentration;

$PO_4$ –Phosphate; POC–Particulate organic carbon flux to the seafloor based on Lutz et al. (2007); SCOC–Sediment-community oxygen consumption; Si–Silicate; SO–salinity; Temp–Bottom temperature; SWS–sea water current velocity; VGPM–monthly surface primary production; VGPM\_svi–Seasonal index of mean monthly surface primary production

and global lithologic class, have confidence intervals that encompass zero across all taxa.

Together, these results define two principal environmental niches within the study area: a cool, oxic, nutrient-poor, higher-salinity water mass that favors most meiofauna, and a warm, low-salinity, nutrient-rich, oxygen-depleted water mass exploited chiefly by Loricifera and, to a lesser extent, Tantulocarida. Food supply and micro-habitat heterogeneity refine these broad patterns but do not overturn the dominance of the chemical and hydrographic gradients.

#### *Effects of grain size on meiofauna abundance*

The compositional data analyses revealed no significant effect of the relative proportion of sand and mud combined versus gravel (ILR1) on the

abundance of any taxon; consequently, ILR1 was not considered further. In contrast, the balance between sand and mud (ILR2) significantly affected meiofaunal abundance (Table 2). Negative coefficients indicate higher abundances in mud-rich sediments, and all estimated effects were consistently negative across taxa. Significant relationships after Benjamini–Hochberg correction were detected for the overall meiofaunal assemblage as well as for Gastrotricha, Tantulocarida, Nematoda, and Ostracoda, whereas the effect for Nauplii was only nominally significant. The consistent direction of the effects suggests a broadly shared preference for fine-grained sediments within the meiofaunal community, although not all taxon-specific responses reached statistical significance.

**Table 2** Effects of sediment composition (ILR2: sand vs. silt/clay) on the abundance of meiofaunal taxa

| Taxon         | $\beta$ (ILR2) | 95% CI           | $p$ (nominal) | $p$ (BH-adj.) |          |
|---------------|----------------|------------------|---------------|---------------|----------|
| Total         | −0.173         | [−0.286, −0.060] | <b>0.0027</b> | <b>0.0107</b> | **/*     |
| Copepoda      | −0.098         | [−0.214, 0.018]  | 0.0982        | 0.1228        | n.s./n.s |
| Nauplii       | −0.094         | [−0.184, −0.004] | <b>0.0400</b> | 0.0666        | */n.s    |
| Kinorhyncha   | −0.116         | [−0.294, 0.062]  | 0.2001        | 0.2223        | n.s./n.s |
| Gastrotricha  | −0.211         | [−0.358, −0.063] | <b>0.0052</b> | <b>0.0130</b> | **/*     |
| Tardigrada    | −0.118         | [−0.548, 0.312]  | 0.5909        | 0.5909        | n.s./n.s |
| Loricifera    | −0.337         | [−0.722, 0.048]  | 0.0860        | 0.1228        | n.s./n.s |
| Tantulocarida | −0.243         | [−0.440, −0.046] | <b>0.0158</b> | <b>0.0316</b> | */*      |
| Nematoda      | −0.180         | [−0.297, −0.064] | <b>0.0024</b> | <b>0.0107</b> | **/*     |
| Ostracoda     | −0.228         | [−0.379, −0.076] | <b>0.0032</b> | <b>0.0107</b> | **/*     |

Negative coefficients indicate higher abundances in mud-rich sediments. Confidence intervals (CI), nominal  $p$ -values and Benjamini–Hochberg (BH) adjusted  $p$ -values are shown. Significance:  $p < 0.05 = *$ ;  $p < 0.01 = **$ ;  $p < 0.001 = ***$ ; n.s. = not significant

## Discussion

This study characterizes the environmental determinants of metazoan meiofaunal densities at abyssal and hadal depths of the KKT and AT, applying different statistical modeling techniques to identify the main predictors and quantify taxon-specific effects. Given the paucity of baseline data for the region, the findings serve as a key reference for predicting benthic community responses to projected climate-driven changes (e.g., shifts in temperature, dissolved oxygen, and primary productivity; Sweetman et al., 2017).

Several environmental predictors of meiofaunal densities were identified (Figs. 3, 4, Table 2). The GLMMs refine, and in part rearrange, the picture that emerged from the Gradient-Forest analysis. The Gradient-Forest analysis ranked surface productivity proxies (i.e., mean chlorophyll-*a* and VGPM) alongside bottom-water temperature and salinity as the most important predictors when their variance was summed across all taxa. In contrast, taxon-specific GLMMs, when accounting for random effects, identified dissolved nutrients (nitrate, phosphate, silicate) and oxygen as consistently significant drivers, whereas the productivity metrics remain significant only for a few taxa such as Kinorhyncha and, to a lesser extent, Copepoda. In addition, grain size seemed to be important, with a higher proportion of mud associated with elevated meiofaunal densities. The importance of hydrographic conditions is supported by both approaches: Bottom-water temperature and salinity rank among the most influential

predictors in the Gradient-Forest analysis and emerge with clear, opposing slopes in the GLMMs. Together, these patterns indicate a differentiation of the deep-sea assemblage along a hydrographic gradient, with taxa associated with cooler, more oxygenated conditions contrasting with those occurring preferentially under warmer, less oxygenated bottom waters.

These divergences highlight the complementary nature of the two methods. Gradient Forest is a multivariate machine-learning approach that integrates responses across multiple taxa to identify environmental gradients associated with the strongest community turnover and compositional change, while accounting for complex, non-linear relationships (Ellis et al., 2012). By optimizing split points to maximize overall variance explained, it emphasizes gradients that structure variation across the entire assemblage, meaning that productivity-related variables such as chlorophyll-*a* and VGPM can appear highly important when they consistently influence many taxa simultaneously, even if their effects are relatively weak or inconsistent at the level of individual groups. In contrast, the GLMMs evaluate each taxon separately and estimate the direction and strength of its response, while accounting for variation among stations and samples (Bolker et al., 2009), thereby focusing on relationships that are robust within taxa once other sources of variability are taken into account. By providing taxon-specific regression coefficients and confidence intervals, GLMMs reveal, which of the shared gradients actually alter local density, once other factors are held constant. As a result, predictors

that consistently affect local abundances, such as oxygen and dissolved nutrients, emerge more clearly as significant in the taxon-specific models, whereas broad-scale productivity proxies, despite their importance for community-wide turnover, show weaker or inconsistent effects on individual taxon abundances. Thus, Gradient Forest captures the broad productivity gradient structuring community-wide compositional turnover, whereas GLMMs identify the nutrient–redox axis as exerting the strongest and most consistent control on taxon-specific abundances.

Ecologically, this implies a two-tier control: surface-derived food availability (chlorophyll-*a*, VGPM, phosphate) establishes the regional capacity to support meiofaunal communities and influences broad-scale distributional patterns and community composition (see also Rex et al., 2006; Wei et al., 2010; Schmidt & Martínez Arbizu, 2015; Ostmann & Martínez Arbizu, 2018), while bottom-water chemistry (oxygen, dissolved nutrients), hydrography and grain size exert stronger, more consistent controls on local population densities and determine which taxa can successfully exploit available resources at a given location (see also Ostmann & Martínez Arbizu, 2018; Wang et al., 2019; Kamenev et al., 2022). Interestingly, salinity and temperature also showed substantial influence, despite their relatively narrow spatial gradients across the KKT and AT (Online Resource 3), suggesting that these variables primarily reflect water-mass characteristics and broader hydrographic regimes rather than acting as direct drivers of meiofaunal densities.

In our study, meiofaunal densities differed markedly between the two trench systems, with the KKT generally supporting higher densities than the AT, although notable variability was observed within each trench (Fig. 2, Online Resource 1). Elevated densities at the trench axes were evident in the KKT, but not in the AT, compared to surrounding abyssal sites. Comparisons across trenches and depth zones (abyssal vs. hadal) indicate that trench axes serve as localized hotspots of meiofaunal abundance. Similar patterns have been reported in the Atacama, Japan, and Tonga trenches, where trench-axis sediments consistently exhibit higher densities relative to adjacent abyssal plains (Danovaro et al., 2002; Itoh et al., 2011; Ichino et al., 2015; Schmidt & Martínez Arbizu, 2015; Schmidt et al., 2019; Shimabukuro et al., 2022). These elevated densities are likely driven by

downslope focusing of organic material along steep trench topography (Itoh et al., 2011; Ichino et al., 2015; Schmidt & Martínez Arbizu, 2015; Schmidt et al., 2019; Shimabukuro et al., 2022).

The lack of a clear trench-axis pattern in the AT may be due to a mass-wasting event, with hadal sediments overlain by a fluid mud layer (Sindlev et al., 2026), resulting in locally depressed abundances (see Preikshardt et al., 2025 for macrofauna). More broadly, trenches are geologically unstable environments subject to frequent seismic activity and submarine turbidites. Such disturbances—likewise inferred for parts of the KKT—act as broad, non-selective suppressors of meiofaunal abundance and likely contribute to the high spatial variability observed in both trench systems (Beliaev & Brueggeman, 1989; Bigham et al., 2021; Kamenev et al., 2022).

Shimabukuro et al. (2022) further demonstrated that, across three hadal trench systems underlying regions of highly variable surface productivity, meiofaunal densities were more closely linked to microbial benthic activity, such as oxygen consumption rates, than to traditional proxies for food availability (e.g., total organic carbon or phytopigments). Similarly, in the KKT this microbial pathway is likely important: the widespread occurrence of hadal cold-seep communities hosting endosymbiotic chemosynthetic bacteria on trench slopes and the trench floor (Fukumori et al., 2019; Peng et al., 2025) indicates that chemosynthetically produced organic carbon can supplement surface-derived material. These findings suggest that microbial biomass represents a critical food source for hadal meiofauna and that variations in food delivery and microbial activity largely drive the spatial heterogeneity of meiofaunal abundance both within and across trench systems.

Several studies have shown that sediment grain size is an important factor in determining the abundance and community structure of meiofauna (Gray, 1981; Carman et al., 1987; Wang et al., 2019; Zhao et al., 2020). Across the assemblage, abundances were higher in mud-rich sediments, with significant responses detected for the total meiofauna as well as several individual taxa, including Nematoda, Gastrotricha, Tantulocarida, and Ostracoda. This pattern is ecologically plausible, as fine-grained sediments, associated with hydrodynamically quiet environments, typically accumulate larger quantities of organic matter and microbial biomass than

coarser sediments, thereby providing enhanced food resources for deposit-feeding meiofauna and their associated microbial food webs (e.g., Diesing et al., 2021). The observed preference for mud-rich habitats therefore likely reflects more indirect effects of food availability and sediment biogeochemistry rather than a direct response to sediment texture itself (but see Tong et al., 2021).

Our analysis revealed distinct responses among meiofaunal taxa to environmental predictors. Specifically, changes in Nematoda and Tardigrada abundances were more strongly linked to mean chlorophyll-*a* concentrations, whereas Ostracoda densities showed a stronger association with VGPM\_svi. Copepoda exhibited significant correlations with salinity (mean sea-surface salinity) and phosphate concentrations. Nauplii partly differed from copepodid and adult copepods in their responses to environmental predictors, particularly with respect to VGPM\_svi and grain size (Fig. 5, Table 2). Notably the threshold for split changes in meiofaunal densities in response to chlorophyll-*a* concentrations was consistent with that reported by Ostmann & Martínez Arbizu (2018), specifically within the 0.6–0.8  $\mu\text{g L}^{-1}$  range.

The separation of nauplii from adult and copepodid Copepoda remains a subject of methodological debate, as nauplii do not constitute an independent taxon. Nevertheless, their retention as a separate ecological category in this study is predicated on the assumption that they possess a distinct morphology, developmental stage, and potentially diverging environmental requirements. The observed differences in responses between nauplii and copepodids/adult copepods in our study further support this view. Similar stage-specific responses to environmental variability have previously been documented in deep-sea meiofaunal studies (e.g., Guidi-Guilvard et al., 2009; Guidi-Guilvard & Dallot, 2014), suggesting that nauplii can provide an ecologically informative operational category in meiofaunal assessments.

As elsewhere, Nematoda dominated the meiofaunal communities in our study (Rosli et al., 2018; Zeppilli et al., 2018). However, some taxa exhibited markedly low densities, which posed challenges for statistical analyses and interpretation. Add to this, the observed high variability in meiofaunal densities at small spatial scales, even within single MUC deployments poses significant challenges in modeling

meiofaunal abundances (see also Uhlenkott et al., 2020). This variability reflects both the natural patchiness of meiofaunal distributions—driven by aggregated reproductive strategies, limited dispersal, and biotic interactions (Findlay, 1981)—and local environmental heterogeneity, including factors such as food availability, disturbance history, sediment grain size, porosity, and organic matter quality (Rosli et al., 2018; Shimabukuro et al., 2022; Kamenev et al., 2022 and citations therein). Small-scale habitat complexity, including microtopography and biogeochemical microniches, can further amplify this patchiness, with meiofaunal densities varying by orders of magnitude over centimeter to meter scales (Gallucci et al., 2009; Leduc et al., 2012). Such fine-scale variability generates overdispersion in predictive models, where observed variance exceeds expectations based on standard probability distributions (Guisan & Thuiller, 2005; Ver Hoef & Boveng, 2007), and can mask broader environmental signals if sampling intensity is insufficient to capture the underlying spatial structure (Warwick et al., 1990).

To mitigate these effects, we applied two different approaches, i.e., Gradient Forests and GLMM with one fixed and a focal parameter. Nevertheless, many taxa had too few observations or very low individual counts (e.g., Acari), which limited model quality and statistical power. The GLMM analyses were based on single-predictor models to avoid overparameterization given the available sample size. Consequently, potential interactions among environmental variables could not be assessed. In addition, environmental predictors were derived from Bio-ORACLE, which provides broad-scale environmental conditions at a coarser spatial resolution than the biologic sampling. As a result, fine-scale environmental heterogeneity may not be fully captured, and some temporal mismatch between environmental and biologic data cannot be excluded.

Future studies based on larger datasets, finer-resolution environmental information, and increased replication across stations will help refine these relationships and improve model robustness. A hierarchical sampling design that explicitly targets the main environmental gradients identified here—productivity, bottom-water temperature, salinity, and oxygen—while ensuring sufficient spatial spacing to capture regional-scale variation (e.g., > 10 km; Levin et al., 2001) would be especially valuable. At the same time,

substantially increasing within-station replication is essential; specifically, increasing the number of cores per station from three to at least five, and ideally ten or more, would allow robust estimation of local variance and better separate true environmental effects from stochastic patchiness (Rose et al., 2005; Gallucci et al., 2009; Uhlenkott et al., 2020). Additionally, modeling taxon presence–absence using binomial GLMMs would complement abundance models and provide insights into environmental controls on occurrence probabilities, which are often more tractable and ecologically interpretable when densities are low or highly variable (Warton et al., 2015; Hui et al., 2013).

## Conclusion

By combining Gradient Forest and GLMM, we disentangled broad-scale vs. taxon-specific drivers of metazoan meiofauna density across the abyssal and hadal zones of the KKT and AT. Both approaches revealed consistent hydrographic influences but differed in the relative weight assigned to surface productivity versus bottom-water chemistry, highlighting the value of complementary statistical frameworks. Overall, the results suggest that surface-derived food availability establishes the regional backdrop for meiofaunal density, while local factors, such as oxygen levels, hydrographic conditions, and disturbance events, modulate community composition at finer scales. In the KKT, chemosynthetically-derived carbon associated with cold-seep communities may supplement surface-derived organic matter, suggesting that hadal food webs are supported by multiple carbon sources and underscoring the complex interplay between regional productivity gradients and local environmental conditions.

Importantly, this study provides critical baseline data for these understudied regions, serving as a reference for assessing benthic responses to projected climate-driven changes, including shifts in temperature, oxygen, and primary productivity (Sweetman et al., 2017). At the same time, the observed high variability in meiofaunal densities at fine spatial scales, even within single MUC deployments, emphasizes the need for comprehensive sampling strategies that capture natural patchiness and environmental heterogeneity. Such variability presents challenges for predictive

modeling by inflating prediction uncertainty and masking broader environmental signals in statistical models, thus highlighting the importance of refining methodologies and expanding spatial coverage. Further, methodological constraints should be acknowledged: GLMMs were restricted to single-predictor structures due to sample size limitations, preventing the evaluation of interactions among environmental drivers, and environmental data from Bio-ORACLE may not fully capture fine-scale variability at the scale of biological sampling. Future studies incorporating larger datasets, higher-resolution environmental data, and improved replication will be essential to resolve these processes and strengthen predictive frameworks for deep-sea ecosystems.

Given their sensitivity to environmental change, meiofauna are valuable indicators of ecological responses to climate-driven and other anthropogenic impacts in deep-sea environments, and integrating such baseline data into predictive frameworks is essential for anticipating cascading effects on biodiversity and ecosystem function (Zeppilli et al., 2015).

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**Data availability** All faunal data analyzed in this study are openly accessible via PANGAEA (Felden et al., 2023) through Bonk et al. (2026) or were obtained from published sources as cited in the manuscript. Environmental layers were obtained from Bio-ORACLE (<http://www.bio-oracle.org/>), as well as from Snelgrove et al. (2018), Straume et al. (2019), and Whitaker et al. (2008).

## Declarations

**Conflict of interest** The authors declare that they have no conflict of interest.

**Ethical approval** No approval of research ethics committees was required to accomplish the goals of this study because experimental work was conducted with unregulated invertebrate species.

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